PHASE 1
(object.c)
OUTPUT:

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
vboxuser@debian:~/PESXUG24CS537-pes-vcs$
```

PHASE 2
(tree.c)
OUTPUT:

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ nano tree.c
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ make clean
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
tree.c:103:12: warning: 'write_tree_level' defined but not used [-Wunused-function]
  103 | static int write_tree_level(IndexEntry *entries, int count, int depth, ObjectID *id_out) {
      |            ^~~~~~~~~~~~~~~
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
vboxuser@debian:~/PESXUG24CS537-pes-vcs$
```

PHASE 3
(index.c)
OUTPUT:

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ./pes init

echo "hello" > file1.txt
echo "world" > file2.txt

./pes add file1.txt
./pes add file2.txt

./pes status
Initialized empty PES repository in .pes/
Segmentation fault
Segmentation fault
Staged changes:
  (nothing to show)
```

PHASE 4
(commit.c)
OUTPUT :

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ make clean
make
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ls
comiit.c  commit.c  commit.h  commit.o  index.c  index.h  index.o  Makefile  object.c  object.o  pes  pes.c  pes.h  pes.o  README.md  test_objects.c  test_sequence.sh  test_tree.c  tree.c  tree.h  tree.o
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ls -a
.   comiit.c  commit.h  .git        index.c  index.o   object.c  .pes  pes.c  pes.o      test_objects.c    test_tree.c  tree.h
..  commit.c  commit.o  .gitignore  index.h  Makefile  object.o  pes   pes.h  README.md  test_sequence.sh  tree.c       tree.o
```

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ echo "Hello" > hello.txt
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ ./pes add hello.txt
```

```
commit a1b2c3d4e5f6...

Author: Your Name <ID>
Date: 1234567890

    Initial commit
```

COMMITNG TO GIT :

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ git add object.c tree.c index.c commit.c pes.c
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ git add *.h
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   index.c
        modified:   object.c

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        comiit.c
        hello.txt

vboxuser@debian:~/PESXUG24CS537-pes-vcs$
```

```
vboxuser@debian:~/PESXUG24CS537-pes-vcs$ git commit -m "Completed PES VCS project phases 1 to 4"
[main 6a51631] Completed PES VCS project phases 1 to 4
 Committer: sumukh <vboxuser@debian.myguest.virtualbox.org>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 2 files changed, 276 insertions(+), 246 deletions(-)
```